CS454 Midterm 1 Part A

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I. Questions

A. General Learning

1. In which of the following machine learning applications there is a supervisor and it tries to estimate a discrete entity from a set of possibilities?

a) Association
b) Classification
c) Regression
d) Unsupervised Learning
e) Reinforcement Learning

2. In which of the following machine learning applications there is no supervisor?

a) Association
b) Classification
c) Regression
d) Unsupervised Learning
e) Reinforcement Learning

B. Classification

Given the following classification dataset $\mathcal{X}_1$. Answer questions 3-5 accordingly.

$$\mathcal{X}_1 = \left[\begin{array}{c|cccc} & x_1 & x_2 & x_3 & C \\ \hline x^{(1)} & 4 & 2 & green & yes \\ x^{(2)} & 6 & 3 & yellow & no \\ x^{(3)} & 7 & 2 & blue & yes \\ x^{(4)} & 3 & 1 & green & no \\ x^{(5)} & 3 & 1 & blue & no \end{array}\right]$$

3. How many classes are there in this dataset?

a) 1
b) 2
c) 3
d) 4
e) 5

4. What is the mean vector of features 1 and 2 ?

a) [4.6 1.8]
b) [1.8 4.6]
c) [6 2]
d) [4 1]
e) [4.4 1.6]

C. Missing Values

$$\mathcal{X}_1 = \left[\begin{array}{c|cccc} & x_1 & x_2 & x_3 & C \\ \hline x^{(1)} & 4 & 2 & green & yes \\ x^{(2)} & 6 & ? & yellow & no \\ x^{(3)} & 7 & 2 & blue & yes \\ x^{(4)} & 3 & 1 & green & no \\ x^{(5)} & 3 & 1 & blue & no \end{array}\right]$$

5. If feature 2 of $x^{(2)}$ is missing, what will be the imputation value for it?

a) 1
b) 1.2
c) 1.3
d) 1.5
e) 2

D. Discrete To Continuous Conversion

6. If you convert feature 3 to continuous using 1-of-L encoding, what can be the code for $x^{(2)}$?

a) 1
b) 0
c) 1 0
d) 0 1 0 0
e) 0 1 0

E. Classification Metrics

Given the dataset in the questions above, the classification decisions for two classifiers are given below. Answer questions 7-8 accordingly.

$$\left[\begin{array}{cc} Classifier_1 & Classifier_2 \\ \hline no & yes \\ no & no \\ yes & yes \\ yes & no \\ yes & yes \end{array}\right]$$

7. If yes is the positive class, then what is FP (False positive) of $Classifier_2$?

a) 0
b) 1
c) 2

d) 3
e) 4

8. If yes is the positive class, then what is the recall of $Classifier_2$?
a) 1 / 1
b) 1 / 2
c) 1 / 3
d) 3 / 4
e) 2 / 3

F. Regression

Given the following regression dataset $\mathcal{X}_2$. Answer question 9 accordingly.

$$\mathcal{X}_2 = \left[\begin{array}{c|ccc} & x_1 & x_2 & r \\ \hline x^{(1)} & 4 & 2 & 4.2 \\ x^{(2)} & 6 & 3 & 3.6 \\ x^{(3)} & 7 & 2 & 2.4 \\ x^{(4)} & 3 & 1 & 6.2 \end{array}\right]$$

9. What is the average regression value of the dataset?
a) 3.8
b) 4.0
c) 4.1
d) 4.3
e) 4.5

G. Regression Metrics

Given the dataset in the questions above, the regression decisions for two regressors are given below. Answer question 10 accordingly.

$$\left[\begin{array}{cc} Regressor_1 & Regressor_2 \\ \hline 4.0 & 4.0 \\ 6.0 & 3.6 \\ 3.0 & 2.2 \\ 7.0 & 6.4 \end{array}\right]$$

10. What is the mean absolute error of Regressor$_1$?
a) 0.7
b) 0.8
c) 0.9
d) 1.0
e) 1.1

H. K class vs. 2 class classification problems

11. If we have a $K = 4$ class classification dataset, how many one-vs-one 2 class problems can be defined for this dataset?
a) 4
b) 5
c) 6
d) 8
e) 16

I. Normal Distribution

12. At which points does the normal distribution take its peak value?
a) mean
b) variance
c) gauss
d) bias

13. When the covariance of two variables are zero and their variances are equal, what is the shape of the isoprobability contour plot?
a) square
b) ellipse
c) circle
d) ellipse with 45 degree

J. Multivariate Classification

14. Which of the following classifiers is the most simple one?
a) Qda
b) Lda
c) Nearest Mean
d) Naive Bayes

15. Which of the following classifiers does not make any dependency assumption between features?
a) Qda
b) Lda
c) Nearest Mean
d) Naive Bayes

K. Experiment Design

16. Which of the following datasets is used to learn the classifier?
a) Train set
b) Test set
c) Validation Set

17. Which of the following datasets is used to get the performance of the classifier in real life?
a) Train set
b) Test set
c) Validation Set

18. What will be the training test percentage if we do 10-fold cross-validation?
a) 10 percent
b) 20 percent
c) 50 percent
d) 80 percent
e) 90 percent

L. Training and Space Complexity

19. Which of the following algorithms has no training complexity?

a) K nearest neighbor
b) Qda
c) Lda
d) Nearest Mean
e) Naive Bayes

20. Which of the following algorithms has the highest space complexity?

a) K nearest neighbor
b) Qda
c) Lda
d) Nearest Mean
e) Naive Bayes

M. Model Selection

21. If the classifier is trained more than it was needed and it memorizes its decisions, then the classifier is ...?

a) Overfitting
b) Underfitting
c) Fitting

22. If the classifier has large training and test errors, then the classifier is ...?

a) Overfitting
b) Underfitting
c) Fitting

N. Feature Selection

23. Which of the following a subset selection technique starts with empty subset of features and adds features one by one?

a) Forward selection
b) Backward selection
c) Floating selection
d) Exhaustive search

24. What is the number of possible subsets of d variables?

a) d
b) d^2
c) 2^d
d) d^3

Given the following error rates for a four feature problem, answer question 25 accordingly. Error Rates: (F_1), 0.23; (F_2), 0.15; (F_3), 0.16; (F_4), 0.24; (F_1, F_2), 0.18; (F_1, F_3), 0.14; (F_1, F_4), 0.20; (F_2, F_3), 0.16; (F_2, F_4), 0.29; (F_3, F_4), 0.22; (F_1, F_2, F_3), 0.19; (F_1, F_2, F_4), 0.25; (F_1, F_3, F_4), 0.24; (F_2, F_3, F_4), 0.26; (F_1, F_2, F_3, F_4), 0.29.

25. What will be the result of the stepwise backward selection?

a) F_2
b) F_3
c) F_1, F_2
d) F_1, F_3
e) F_1, F_2, F_3

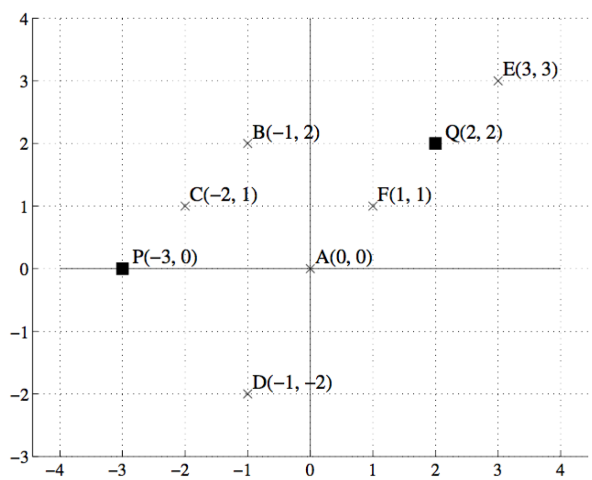
O. Feature Extraction

26. If the eigenvalues of the covariance matrix are $\lambda_1 = 2$, $\lambda_2 = 1$, $\lambda_3 = 0.5$, $\lambda_4 = 0$, and $\lambda_5 = 0$; then with proportion of variance explained 90 percent, what will be the reduced dimensionality?

a) 1
b) 2
c) 3
d) 4
e) 5

P. Clustering

Suppose we are given the following dataset. In the following questions 51-54, we will do k-means clustering to cluster the points A, B ... F into 2 clusters. The current cluster centers are P and Q.



27. Which points are assigned to cluster center Q?

a) E, F
b) E
c) A, F, E
d) A, B, F, E
e) F

28. What does cluster center Q get updated to?

a) 2, 2
b) 3, 3
c) 4 / 3, 4 / 3
d) 3 / 4, 6 / 4
e) 1, 1

Q. K-NN

Assume you wish to use the K-nearest neighbors algorithm on this dataset.

A	B	C	D	Output
T	T	T	T	F
F	F	T	F	T
F	T	F	T	F
T	T	F	F	T
T	F	T	T	F
T	T	F	T	?

29. Which class will be assigned to the test instance (last instance) if k = 1?
a) T
b) F

30. Which class will be assigned to the test instance (last instance) if k = 3?
a) T
b) F